Frozen Visual Representations for Relational Outfit Compatibility: A Leakage-Audited Cross-Dataset Study

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Abstract

We compare seven released frozen visual representations for outfit compatibility using a shared pairwise logistic classifier and principal-component analysis fitted only on training items. We develop the comparison on leakage-audited Polyvore-D-Clean, replicate it prospectively on the frozen IQON3000-Clean protocol, and evaluate transfer to unchanged A100 questions based on human and expert judgments. On both corpora, the evaluated contrastive vision-language model (VLM) representations outperform ResNet50 and generic DINOv3 on compatibility prediction and fill-in-the-blank completion. A100 supports this broad ordering, although it does not reliably distinguish closely ranked models. The tier-level result persists after exclusion of the two checkpoints with the greatest fashion-data provenance uncertainty. The benefit of fashion specialization depends on the checkpoint and training recipe. Comparisons with the same checkpoints in LookBench show that retrieval ranks do not reliably predict compatibility ranks. The principal conclusion remains stable across construction seeds, feature dimensionality, nonlinear downstream learning, provenance restrictions, label availability, and dependence-aware sensitivity analyses. Calibration, efficiency, and provenance provide complementary model-selection criteria. RepBench-Fashion provides the protocol builders, evaluation code, and statistical analyses needed to reproduce these comparisons.

Keywords: outfit compatibility, representation benchmarking, vision-language models, leakage-audited evaluation, calibration, computational efficiency

1 Introduction

Fashion recommendation includes several related but non-equivalent problems: retrieving an item similar to a query, predicting a user's preference, completing a partial outfit, and judging whether a set of garments is mutually compatible. Recent surveys describe a field that increasingly combines visual, textual, and behavioural signals, while also noting persistent fragmentation in datasets, tasks, and evaluation practice [1, 2]. Item retrieval asks for proximity to an anchor; outfit compatibility asks whether different garments form a plausible whole. A representation that is excellent for finding visually similar products need not expose the cues required for colour coordination, formality, season, silhouette, or balance. These tasks let us study how visual representations transfer to judgments about relations between items.

The standard Polyvore literature established compatibility prediction (CP) and fill-in-the-blank (FITB) completion as widely used evaluations [3–5]. It also encouraged increasingly expressive task-specific architectures: sequence models, type-conditioned metrics, graph reasoning, attention, set models, and transformers. With supervised convolutional networks, self-supervised vision transformers, and contrastive vision-language models (VLMs) now available, we ask how much compatibility information a frozen image representation provides when the downstream learner is held fixed.

Answering that question requires more than replacing one backbone in an existing pipeline. First, outfit datasets contain repeated items and duplicate images. Random or legacy splits can therefore allow the same product evidence to cross training and test boundaries, a concrete form of leakage [6, 7]. Second, synthetic negatives and FITB distractors are part of the measurement instrument: changing them can change both absolute scores and rankings. Third, fashion-specialized checkpoints may have incomplete data provenance, especially when their upstream data are web-scale or proprietary. Fourth, point estimates alone obscure paired uncertainty and the dependence induced by shared outfits, items, or users. Finally, predictive quality is only one deployment criterion; probability calibration and representation cost can favour different choices.

We make the released representation checkpoint the experimental unit. Seven frozen image representations receive exactly the same downstream transformation: a scaler and randomized principal-component analysis (PCA) fitted on training items only, an unordered pair construction, mean pooling over all item pairs, and a regularized logistic classifier. Holding the relational learner, labels, candidate sets, and selection logic fixed prevents representation choice from being confounded with simultaneous changes to architecture, modality, loss, personalization, or question construction. The roster spans an ImageNet-supervised ResNet50 [8, 9]; the self-supervised DINOv3 ViT-L/16 [10–12]; generic CLIP and SigLIP2 VLMs [13–15]; FashionCLIP [16]; and the fashion-retrieval-oriented Marqo-FashionSigLIP and GR-Lite [17–19]. The study's scope is controlled representation transfer for relational outfit compatibility.

The protocol audit detected cross-split item sharing and exact-image duplicates hidden behind different identifiers, rebuilt Polyvore by item/image-connected components, and identified an IQON component containing 98.08% of valid outfits. The IQON construction makes the trade-off between split integrity and source-population coverage explicit: the retained subset preserves nearly all users and the category mixture, while the audit identifies measurable shifts in outfit length and item repetition. Generated CP negatives and FITB distractors are checked for collisions, and a historical-to-clean question bridge separates the effect of regenerated questions from the effect of split cleaning. Polyvore-D-Clean provides development-influenced generalization evidence; the frozen IQON3000-Clean protocol, constructed prospectively from the larger IQON3000 corpus [20], provides replication within a second leakage-audited target. A100 adds unchanged external evidence grounded in expert and human judgments [21]; no A100 label or candidate enters training, calibration, or selection. Additional analyses cover an image-only OutfitTransformer adaptation [22], five construction seeds, feature width, nonlinear downstream learning, compatibility-label efficiency, calibration, compute, checkpoint provenance, failure consensus, and exact-checkpoint LookBench retrieval ranks.

We organize the study around four gaps. Contemporary frozen checkpoints lack a systematic compatibility comparison that holds downstream capacity and question construction constant across leakage-audited corpora (G1). It is also unclear how frozen-transfer results relate to contextual learning, fashion specialization, retrieval quality, and external human and expert judgments (G2). Model comparisons need to account for calibration, cost, and checkpoint provenance alongside discrimination (G3). Finally, their conclusions need testing against duplicate leakage, question construction, subset selection, dependence, feature width, learner choice, and the availability of labelled outfits (G4).

The corresponding objectives are to compare and replicate the shared-probe ordering (O1), assess how far it extends to other tasks and models (O2), compare practical model-selection criteria (O3), and test sensitivity to these validity threats (O4). We contribute a controlled seven-checkpoint benchmark with prospective IQON3000-Clean

replication (C1). Untouched A100 questions, a contextual baseline, six predefined contrasts, and exact-checkpoint retrieval comparisons establish the scope of the findings (C2). We report discrimination together with benchmark calibration, computational cost, provenance, and statistical uncertainty (C3). The protocol builders and analysis code cover duplicate connectivity, generated-question collisions, historical question regeneration, IQON retention and dependence, construction seeds, feature width, learner choice, label availability, and frozen artifact traceability (C4). Section 3 connects these objectives to the research questions and evidence.

2 Related work

2.1 Fashion compatibility, recommendation, and datasets

Early image-based recommendation work learned notions of style and substitution from co-occurrence and preference data [23, 24]. Polyvore-based research then formalized set-level compatibility and FITB completion. The bidirectional-LSTM formulation of Han et al. [3] treated outfits as ordered sequences, while type-aware embeddings [4], mixed-category metric learning [25], and cross-category compatibility learning [26] made item type explicit. Context-aware models reason over an item relative to the remaining outfit [5]; set-based models remove arbitrary sequence assumptions [27]; and attentive knowledge distillation transfers structure from more elaborate teachers [28]. OutfitTransformer provides a recent contextual set representation for recommendation and compatibility [22].

Compatibility has also been approached through capsule wardrobes [29], personalized outfit generation [30], personalized pairwise ranking [31], and multi-layer comparison networks that diagnose incompatible pieces [32]. Attribute-aware and explanatory methods explain compatibility scores [33–38]; a recent review organizes this explainable-compatibility literature [39]. Self-supervised visual attribute learning further shows that compatibility-relevant signals can be shaped without direct labels [40]. These approaches vary in modality, personalization, objective, and downstream capacity. We hold the compatibility learner constant to measure the effect of the representation supplied to it.

Polyvore's catalog-like images and repeated products made it influential for controlled compatibility tasks, while IQON3000 added a larger user-curated collection with user and stylistic context [20]. Broader fashion datasets such as DeepFashion support recognition and retrieval rather than outfit compatibility per se [41]; Fashion200K supports language-aware fashion concepts [42]; and Street2Shop targets cross-domain product retrieval [43]. These task differences motivate caution when transferring conclusions from item retrieval to relational compatibility.

A100 was introduced specifically to assess aesthetic ability beyond synthetic Polyvore evaluation [21]. Its released five-choice *Liberalism Aesthetic Test* (LAT) and *Academicism Aesthetic Test* (AAT) probe whole-outfit aesthetic choice; AAT records are grouped by colour, style, occasion, season, material, and balance. We use these human and expert judgments to assess external transfer and report uncertainty, including sensitivity to two released LAT label interpretations. A subsequent higher-order network study uses A100 within a task-specific compatibility and cross-selling system [44].

2.2 Generic and domain-specialized visual representations

The transferability of representations is not guaranteed by source-task accuracy. ImageNet model quality often correlates with transfer, but task and protocol can alter the relationship [45]. Similar variation occurs among self-supervised objectives [46]. Broad suites such as VTAB, ELEVATER, and WILDS expose the importance of multiple target tasks and distribution shift [47–49]. Recent e-commerce embedding comparisons likewise evaluate foundation-model transfer, tuning, retrieval, classification, and practical cost [50]. In applied computer vision, systematic comparison of large and small MLLMs across capability categories and application settings has shown how benchmark design can expose capability boundaries and deployment trade-offs without making architecture invention the experimental question

[51]. RepBench-Fashion brings this controlled-measurement perspective to a relational visual target.

Vision Transformers [52] enabled scalable visual encoders, while CLIP aligned image and language representations through contrastive web supervision [13]. ALIGN demonstrated similar scaling with noisy image-text data [53]; LAION-5B later provided an open web-scale resource for such training [54]. SigLIP replaced global softmax normalization with independent sigmoid losses [14], and SigLIP2 combined that foundation with multilingual, self-supervised, captioning, and curation objectives [15]. In parallel, DINO and DINOv2 established strong label-free ViT features [10, 11]; DINOv3 scaled this family and introduced mechanisms for stable dense representations [12].

Fashion-specific adaptation can encode product vocabulary and domain cues. FashionCLIP fine-tunes a CLIP-style model on more than 800,000 Farfetch product image-text pairs and reports gains on fashion retrieval, classification, and grounding [16]. Generalized Contrastive Learning incorporates graded rankings into a retrieval representation [17]; Marqo-FashionSigLIP is a released fashion checkpoint associated with that ecosystem [18]. GR-Lite is the open fashion retrieval model released with LookBench [19]. Recent retrieval audits and specialization efforts further show that benchmark composition and fine-tuning data affect fashion retrieval conclusions [50, 55]. Domain specialization might help compatibility, but it might instead emphasize item similarity or reflect a particular training recipe. Direct controlled comparison is therefore necessary.

2.3 Closest contemporary systems and scope

Recent work defines the closest boundary to our contribution. VICTOR combines fashion-specific contrastive pretraining with a multitask Transformer for incompatibility detection [56]; OutfitTransformer learns contextual set representations [22]; HAT incorporates personal history [57]; and MEDAL learns task-specific distilled embedding spaces [58]. Text2Outfit studies multimodal large-model outfit generation [59], while recent systems fine-tune multimodal models or pretrain cross-item relations for product bundling [60, 61]. LMLMO combines generated fashion cues with BEiT3 compatibility representations [62], and FashionStylist adds expert-knowledge-enhanced data and multimodal styling evaluations [63]. Their results concern end-to-end or task-specific systems, whose training and evaluation differ from a shared frozen-checkpoint comparison.

Prior work has therefore incorporated or adapted modern representations within compatibility systems. The distinct contribution here is to determine which transfer conclusions remain defensible when the relational probe and evaluation logic are fixed across item- and exact-image-disjoint corpora, prospective replication, external corroboration, provenance, protocol calibration, efficiency, and construction stability. Table 1 states this boundary without treating different splits, modalities, negative generators, personalization targets, or training objectives as directly comparable leaderboards. Published VICTOR, HAT, MEDAL, HON, Text2Outfit, and related scores are consequently discussed but not numerically compared with ours.

2.4 Calibration, efficiency, and reproducible comparison

ROC–AUC measures discrimination, not whether predicted probabilities correspond to empirical frequencies. The Brier score is a proper squared-probability loss [64]; log loss offers a complementary proper scoring rule; and expected calibration error (ECE) summarizes reliability-bin discrepancies, although its value depends on binning. Calibration studies show that modern neural systems can be accurate yet miscalibrated and that calibration assessment itself requires care [65–69]. Under distribution shift, uncertainty can degrade independently of accuracy [70]. We report discrimination, proper scores, and ECE sensitivity separately.

Efficiency has analogous measurement problems. Parameter count, operations, latency, throughput, memory, and end-to-end extraction time answer different questions; “efficient” without a declared measure is underspecified [71]. Hardware and software stacks affect results [72], while systematic compute and carbon reporting

Table 1 Scope of the closest contemporary work. "External" denotes human/expert-grounded evidence outside the training corpora.

Study	Model treatment and evaluation	Additional axes	Boundary relative to this study
VICTOR [56]	Fashion-pretrained backbones and multitask Transformer; Polyvore, MISFIT	Encoder cost	Task-specific incompatibility detection
OutfitTransformer [22]	End-to-end contextual set model; Polyvore	–	Contextual architecture, not frozen transfer
HAT [57]	End-to-end history-aware Transformer; Polyvore, IQON	Personalization	User-history recommendation
MEDAL [58]	Distilled task-specific meta-spaces; Polyvore	–	Learned spaces, not a common frozen probe
LMLMO [62]	Generated cues with BEiT3; FashionVC, EVALUATION3	Multimodal reasoning	Pair compatibility under a different protocol
HON/A100 [44]	End-to-end higher-order network; Polyvore, A100	External aesthetics	Task-specific compatibility and cross-selling
E-commerce benchmark [50]	Frozen and tuned embeddings; e-commerce item data	Retrieval; classification; cost	Item-level representation benchmark
LookBench [19]	Frozen and tuned retrieval models; LookBench	Human labels; retrieval; audit; cost	Live fine-grained item retrieval
RepBench-Fashion (ours)	Seven frozen checkpoints and contextual baseline; Polyvore, IQON, A100	Leakage; retrieval; calibration; cost; statistics; provenance	Common relational probe with prospective replication

improves accountability [73–75]. We report same-device measurements and Pareto frontiers, but avoid extrapolating a single GPU result to all deployments.

Finally, model comparisons require paired analysis and disciplined multiplicity. Bootstrap methods provide distribution-light interval estimates [76]; repeated training and benchmark construction introduce additional variance that should be reported [77]. Multiple-dataset comparisons and resampled estimates require care because observations and folds are dependent [78, 79]. We use group-aware paired resampling, exact McNemar tests for paired FITB decisions where applicable, and Holm's sequential correction for predefined contrast families [80]. Protocol hashes, manifests, and audit artifacts follow reproducibility recommendations to make the reported result traceable [81].

3 Research questions and study design

We froze the benchmark before the final IQON and A100 evaluations to address four research questions.

RQ1: Under an identical frozen-feature relational probe, how do the seven released checkpoints compare on item- and exact-image-disjoint Polyvore, and does the broad hierarchy prospectively replicate on IQON3000-Clean?
RQ2: How far do those conclusions extend to A100 human/expert-grounded questions, task-specific contextual learning, fashion-specialization contrasts, and retrieval-versus-compatibility ranking?
RQ3: Do discrimination, benchmark calibration, measured efficiency, and checkpoint provenance favour the same representation?
RQ4: Which conclusions survive audits of leakage and question validity, historical question regeneration, IQON retention and dependence, construction seeds, feature width, learner choice,

compatibility-label availability, outfit length, and failure consensus?

The experimental unit is a frozen representation checkpoint. Images are encoded once. All representations receive the same downstream feature construction, training labels, candidate sets, and validation criterion. Hyperparameters are selected separately per representation using validation CP ROC–AUC, never test or A100 outcomes. Polyvore-D-Clean influenced development choices; its results therefore establish controlled in-domain evidence, not a prospective blind test. The IQON protocol and evaluation path were frozen subsequently and are treated as replication. We distinguish development evidence from replication throughout the analysis.

Six contrasts were specified before final test evaluation, each tied to a distinct scientific or model-selection question:

1. DINOv3 versus ResNet50 tests generic modern self-supervision against the conventional supervised CNN baseline.
2. GR-Lite versus DINOv3 tests fashion-retrieval-oriented specialization relative to the generic DINO-family reference.
3. FashionCLIP versus CLIP tests fashion-domain specialization relative to its generic contrastive VLM family.
4. Marqo-FashionSigLIP versus SigLIP2 tests fashion-oriented specialization relative to a generic sigmoid-loss VLM family.
5. The validation-selected best fashion-specialized versus validation-selected best generic checkpoint tests the practical family-level question of whether specialization improves checkpoint selection.
6. The validation-selected best overall checkpoint versus ResNet50 quantifies the practical gain over the conventional baseline.

Final test outcomes were not used to select arbitrary pairwise comparisons. Holm correction controls this predefined contrast family within each dataset–metric family. Restricting inference to these contrasts limits multiple testing and post hoc interpretation. Retention, item-component, and collision analyses added after the audit are explicitly labelled as post-audit sensitivity analyses rather than predefined hypothesis tests.

Table 2 links the research questions to their objectives and evidence.

4 Datasets and leakage-audited protocol construction

Table 3 summarizes the four protocols. CP is a balanced binary task: each observed outfit is a positive, and its matched negative replaces *every* item with a different same-category item from the same split while retaining outfit length and category sequence. FITB supplies one partial outfit and four candidates, exactly one of which is the held-out item; the remaining three are distinct same-category, within-split distractors. Candidate positions are balanced. These controlled corruptions reduce category and length shortcuts but remain synthetic and should not be interpreted as the population distribution of incompatible outfits.

Question construction explicitly rejects any CP item set already observed as a positive, duplicate generated negatives, any FITB distractor completion already observed as a positive, and duplicated candidates within a question. A frozen-manifest audit additionally found no candidate duplication by exact decoded pixels. Across all six clean splits, the audit found zero CP positive collisions by item or exact-image set and zero FITB positive-completion collisions; the largest cross-question reuse count for a distractor was one on Polyvore and two on IQON. Cross-question reuse is allowed, but no within-question duplication is. Section 7.1 and the supplement report these validation counts.

4.1 Polyvore-D-Clean

Starting from the pooled historical Polyvore-D positives, eight exact duplicate outfits were removed. We formed a graph whose nodes were outfits and whose edges indicated a shared item identifier, an identical encoded image, or identical decoded pixels. There were 120 exact-image duplicate groups. Connected components, rather than individual outfits, were assigned deterministically to train, validation, or test, targeting the historical split sizes. The final split contains 16,991/2,999/15,142 positive outfits and 68,166/13,780/70,839 unique items. Audits

Table 2 Research questions, objectives, and evidence. Each row links a gap to its objective and contribution; Section 8 gives the corresponding explicit answer.

Gap → objective	Research question	Contribution	Main evidence
G1 Controlled-transfer gap → O1 compare and replicate	RQ1 How do frozen checkpoints compare on leakage-audited Polyvore, and does the hierarchy replicate on the frozen IQON protocol?	C1 Controlled seven-checkpoint benchmark with family-level replication under provenance restrictions	Table 5; Fig. 1; Secs. 7.2, 7.3
G2 Model/construct boundaries → O2 assess scope	RQ2 Does the result extend to A100, contextual learning, specialization, and retrieval ranking?	C2 External evaluation and assessment of scope	Figs. 2, 3; Secs. 7.5, 7.7
G3 Single-axis selection → O3 compare operational criteria	RQ3 Do discrimination, benchmark calibration, cost, and provenance favour the same checkpoint?	C3 Multi-criterion model-selection audit	Table 6; Figs. 4, 5
G4 Validity risks → O4 stress-test conclusions	RQ4 Which conclusions survive leakage, construction, selection, dependence, and probe-sensitivity audits?	C4 Leakage-audited protocols and reproducible validity/robustness suite	Sections 7.1, 7.10; Online Resource 1

Table 3 Evaluation protocols and evidential roles. Counts are positive outfits; CP adds one matched synthetic negative per positive. A100 contains fixed questions rather than train/validation/test outfits.

Protocol	Role	Train / val. / test	Split separation	Qualification
Historical Polyvore-D	Comparison	16,995 / 3,000 / 15,145	Neither item nor image	Packaged negatives; known overlap
Polyvore-D-Clean	Development	16,991 / 2,999 / 15,142	Item and exact image	Development informed
IQON3000-Clean	Replication	86,800 / 12,400 / 24,800	Item and exact image	Audited selection effect; not user-disjoint
A100	External	– / – / 200	Exact-image audit	100 LAT + 100 AAT; wide uncertainty

report zero cross-split item-identifier, encoded-file, decoded-pixel, or outfit overlap.

Negatives and FITB candidates were then reconstructed inside each split using category and outfit-length constraints. This order is important: generating corruptions before splitting could reintroduce cross-split evidence. Historical Polyvore-D retains its original 16,995/3,000/15,145 outfits and packaged candidates solely for comparison with prior work. It is not used to support leakage-audited claims.

4.2 IQON3000-Clean

The source archive contains 308,747 JSON records. After requiring at least two distinct items, 297,916 valid outfits from 3,568 users and 712,170 unique source items remained; 610 duplicate outfits were detected. Direct component splitting was infeasible because one item/image-connected component contained 292,197 outfits (98.08% of the valid corpus). We therefore used deterministic balanced greedy hypergraph packing. Item identifiers and exact encoded/decoded image groups were indivisible; outfits that would connect split boundaries were discarded. Exact targets of 86,800/12,400/24,800 positive outfits were met, retaining 41.62% of valid source outfits. The test set yields 49,600 CP examples and 24,800 FITB questions.

The retained protocol has zero cross-split identifier, exact encoded-image, decoded-pixel, and outfit overlap. It is not user-disjoint: the giant connectivity structure and retention target make item/image disjointness the enforced constraint, while users may occur in multiple splits. The retained-versus-excluded audit preserves 98.74% of source users and finds very small category-distribution divergence (Jensen–Shannon divergence 0.0016 bits). It also measures the selection effect: retained outfits are shorter (mean 5.28 versus 6.47 items) and contain less frequently repeated items (mean source frequency 8.73 versus 13.44). The retained subset therefore preserves user coverage and category proportions but underrepresents longer outfits and frequently reused items. IQON confidence intervals primarily resample user clusters; an item-connected-component sensitivity is added because user clustering does not capture all graph dependence. IQON3000-Clean is a prospective replication within the leakage-audited IQON target, not a new-user generalization benchmark.

4.3 A100 external evaluation

A100 is left unchanged and evaluated exactly as its released five-choice records: the probe scores the completed outfit produced by appending each candidate to the supplied partial outfit and selects the highest score. LAT contains 847 image-reference instances resolving to 680 referenced files and 680 unique exact images; AAT contains 983 instances resolving to 983 files and 977 unique exact images. Sixty-five exact images occur in both tasks, giving 1,663 referenced files but 1,592 globally unique exact images. None overlaps either clean training corpus by encoded bytes or decoded pixels. For LAT, the primary target is the majority label $\arg\max(\texttt{gt_distribution})$; mean LAT (mLAT) credits the selected option's vote share. The archive's separate `gt` field disagrees on questions 92, 93, 94, 96, 98, 99, and 100, so archive-gt accuracy is a sensitivity analysis. AAT questions are partitioned as colour (1–20), style (21–52), occasion (53–67), season (68–79), material (80–91), and balance (92–100). Nothing is fitted, tuned, calibrated, or selected on A100.

5 Frozen representation benchmark

Table 4 defines the roster. Each checkpoint is image-only at evaluation time; text is used only historically in VLM pretraining. Official preprocessing is retained, including resolution, interpolation, and normalization. ResNet50 uses the ImageNet-1K V2 weights and a 2,048-dimensional pooled feature. DINOv3 uses the normalized ViT-L/16 class-token representation. CLIP and SigLIP2 use their projected image embeddings. FashionCLIP uses its 512-dimensional projected embedding. Marqo-FashionSigLIP uses the normalized OpenCLIP image output, and GR-Lite uses its released 1,024-dimensional normalized class-token output.

The roster was selected to represent distinct pretraining regimes rather than a sequence of parameter-matched ablations. Parameter count and resolution consequently differ. This is intentional for model-choice measurement, but it prevents causal attribution of a result to one factor such as loss, data, or architecture. Exact revisions, preprocessing, licenses, and provenance notes are recorded because checkpoint identity is part of the intervention.

For an outfit $O = \{x_1, \ldots, x_n\}$, a frozen encoder gives $e_i = f(x_i)$. A scaler and randomized PCA are fitted only to distinct training-item embeddings and applied to validation and test items. Unless otherwise stated, PCA retains 256 dimensions and the result is ℓ_2 normalized to z_i. Each unordered item pair is represented as

$$q_{ij} = [|z_i - z_j| \,;\, z_i \odot z_j], \qquad i < j, \tag{1}$$

and outfit features are their mean,

$$\phi(O) = \frac{2}{n(n-1)} \sum_{i<j} q_{ij}. \tag{2}$$

This 512-dimensional construction is invariant to item order and gives every pair equal weight. A second StandardScaler is fitted to training CP examples only. Logistic regression predicts

$$p(y = 1 \mid O) = \sigma(w^\top \tilde{\phi}(O) + b). \tag{3}$$

The fixed-capacity learner exposes coordinate-wise distance and agreement while keeping the

Table 4 Frozen image representations. Revisions identify the recorded checkpoint commit; provenance summaries reflect the documentation available for each release.

Representation	Family / main pretraining	`Revision`	Input / dim.	Provenance summary
ResNet50	supervised CNN; ImageNet-1K	`11ad3fa6`	224 / 2,048	documented ImageNet-1K
DINOv3 ViT-L/16	self-supervised ViT; LVD-1689M	`ea8dc286`	224 / 1,024	documented LVD-1689M
CLIP ViT-L/14@336	generic contrastive VLM; 400M web pairs	`ce19dc91`	336 / 768	web-scale data
SigLIP2-B/16@384	sigmoid-loss VLM; WebLI mixture	`f775b65a`	384 / 768	web-scale data
FashionCLIP 2.0	fashion contrastive VLM; LAION + Farfetch	`7e3ba62c`	224 / 512	documented web/fashion data
Marqo-FashionSigLIP	fashion sigmoid-loss VLM; SigLIP + fashion GCL	`c56244cc`	224 / 768	fashion data partly disclosed
GR-Lite	fashion retrieval ViT; DINOv3 + fashion data	`a8057f8a`	336 / 1,024	fashion data partly disclosed

visual encoder and relational evaluation protocol constant across representations.

6 Experimental setup

The logistic classifier uses L-BFGS for at most 2,000 iterations, no class weighting, and $C \in \{0.01, 0.1, 1, 10, 100, 1000\}$. Validation CP ROC–AUC is the sole selection criterion; the classification threshold is 0.5. CP reports ROC–AUC as the primary metric, with PR–AUC, accuracy, balanced accuracy, Brier score, log loss, and equal-width ECE. FITB selects the candidate producing the highest completed-outfit compatibility score and reports accuracy; its scores are not interpreted as normalized candidate probabilities.

Uncertainty uses 2,000 paired bootstrap replicates. Polyvore resamples connected-component groups; IQON primarily resamples users; A100 resamples questions. Because IQON test outfits remain dependent through an item graph, a conservative sensitivity instead resamples its 9,231 test item-connected components; the largest contains 11,547 of 24,800 positive outfits. The six predefined contrasts are corrected by Holm's method within each dataset–metric family. CP interval estimates are paired and group-aware. FITB paired differences additionally use exact McNemar tests where the protocol permits, with bootstrap intervals for effect size. Rank replication is summarized by Spearman's ρ and Kendall's τ, treated descriptively given seven models.

Robustness checks were frozen alongside the main analysis, except the explicitly labelled post-audit IQON retention, item-component bootstrap, and question-collision diagnostics added in response to validity review. Construction seeds 42–46 hold item partitions fixed but regenerate same-category, within-split CP negatives and FITB distractors, then refit affected scalers and classifiers. Together with the collision audit and historical-to-clean question bridge, this tests whether the conclusion depends on one generated-question seed, exact false-negative collisions, or the change from packaged to reconstructed questions. Representation width is tested by replacing PCA-256 with native-dimensional features. Learner dependence is tested on IQON using XGBoost over the same outfit features. The compatibility-label analysis uses deterministic 10%, 25%, 50%, and 100% subsets of labelled training outfits; at every fraction, unsupervised PCA still has access to embeddings for all train-split items, while the outfit-feature scaler and classifier are refitted only on the selected labelled outfits. This measures labelled-outfit demand without changing the frozen encoder. Calibration uses 10, 15, and 20 equal-width bins. The provenance-restricted sensitivity removes Marqo-FashionSigLIP and GR-Lite, the two checkpoints

with the greatest fashion-data provenance uncertainty. Failure and outfit-length analyses are descriptive.

The image-only OutfitTransformer adaptation projects cached ResNet50 features to 64 dimensions, prepends an outfit token, and applies six Transformer layers with 16 heads, a 256-unit feed-forward block, dropout 0.1, and no positional encoding. A two-layer binary head is optimized with focal binary cross-entropy ($\gamma = 2$, $\alpha = 0.25$), AdamW (learning rate 10^{-5}, weight decay 10^{-4}), batch size 50, for at most 30 epochs with validation-AUC early stopping (patience 5). Five training seeds (42–46) are reported. This tests whether task-specific contextual learning from a conventional visual backbone can compensate for weaker frozen item features. It is an adaptation to the image-only protocol, not an exact reproduction of the canonical multimodal OutfitTransformer.

Efficiency measurements use one NVIDIA GeForce RTX 5070 Ti (16 GB). Repeated batch-64 inference follows two warm-up and five measured repeats; PyTorch FlopCounterMode estimates operations. End-to-end IQON extraction uses batch 32 across 434,089 retained item images. PCA and classifier timings are reported separately. Hardware-specific values support within-device comparisons only.

7 Results

7.1 Protocol and construction audit (RQ4)

Exact split audits found no cross-split item identifier, encoded-file, decoded-pixel, or outfit overlap in either clean corpus. The construction audit described in Section 4 also found no exact positive–negative collision or duplicated FITB candidate. These checks establish the audited item-, exact-image-, outfit-, and generated-question controls; their scope is defined in Section 9.

The IQON retention audit shows the cost of this guarantee. The protocol retains 41.62% of valid outfits while preserving almost all users and the category mixture, and it identifies that retained outfits are shorter and contain less-repeated items. Moreover, 46.56% of retained test positives remain in one test item-connected component. Primary user-cluster intervals therefore cannot eliminate all item-graph dependence. Component-cluster sensitivity is deliberately conservative: all six predefined IQON contrast directions remain unchanged, including DINOv3–ResNet50 CP 95% CI [0.0517, 0.0651], Marqo–CLIP [0.0140, 0.0237], and the small adverse FashionCLIP–CLIP contrast [−0.0105, −0.0003]. This supports the reported hierarchy while leaving retained-subset generalization as a bounded limitation.

7.2 Polyvore-D-Clean (RQ1)

Table 5 shows a large separation between conventional supervised features and the evaluated VLM representations. ResNet50 reaches 0.6938 CP ROC–AUC and 0.4063 FITB accuracy. DINOv3 improves these by 0.0885 AUC (95% CI 0.0817–0.0952) and 0.0985 FITB (0.0882–0.1088); both predefined comparisons remain supported after Holm correction. GR-Lite adds 0.0455 AUC (0.0399–0.0511) and 0.0525 FITB (0.0425–0.0619) over its generic DINOv3 base family. The three strongest CP point estimates (CLIP, SigLIP2, and Marqo-FashionSigLIP) are within 0.0014 AUC of one another.

Specialization does not yield a uniform advantage. FashionCLIP trails CLIP by 0.0193 AUC (95% CI −0.0232 to −0.0156) and 0.0325 FITB (−0.0405 to −0.0238), with both contrasts supported after correction. Marqo-FashionSigLIP differs from SigLIP2 by only 0.0005 AUC (−0.0026 to 0.0036) and −0.0049 FITB (−0.0131 to 0.0027); neither result supports a Polyvore advantage. The validation-selected best fashion-versus-generic contrast is likewise near zero. The strongest overall system improves over ResNet50 by 0.1959 AUC and 0.2416 FITB. The Polyvore result therefore establishes an advantage for the evaluated VLM tier while showing that domain specialization alone is not sufficient.

Historical Polyvore-D produces the same broad ordering but different absolute values: ResNet50 0.6894/0.4013, DINOv3 0.7605/0.4722, CLIP 0.8712/0.6072, SigLIP2 0.8729/0.6224, FashionCLIP 0.8567/0.5947, Marqo 0.8772/0.6199, and GR-Lite 0.8203/0.5444 for CP/FITB. To separate split cleaning from question construction, we also evaluated the

Table 5 Primary test results on the two item- and exact-image-disjoint protocols. Brackets are 95% group-aware bootstrap confidence intervals. FITB chance accuracy is 0.25. Bold denotes the largest point estimate within a dataset and metric, not significance against every alternative.

Dataset	Frozen representation	CP ROC–AUC [95% CI]	FITB accuracy [95% CI]
Polyvore-D-Clean	ResNet50	.6938 [.6881, .6997]	.4063 [.3981, .4145]
	DINOv3 ViT-L/16	.7822 [.7772, .7873]	.5048 [.4972, .5134]
	CLIP ViT-L/14@336	.8883 [.8843, .8919]	.6500 [.6424, .6576]
	SigLIP2-B/16@384	.8892 [.8857, .8927]	**.6528 [.6454, .6602]**
	FashionCLIP 2.0	.8690 [.8648, .8730]	.6175 [.6099, .6250]
	Marqo-FashionSigLIP	**.8897 [.8859, .8931]**	.6479 [.6408, .6557]
	GR-Lite	.8277 [.8232, .8323]	.5573 [.5498, .5657]
IQON3000-Clean	ResNet50	.5852 [.5798, .5909]	.3166 [.3097, .3233]
	DINOv3 ViT-L/16	.6419 [.6362, .6475]	.3660 [.3587, .3739]
	CLIP ViT-L/14@336	.7042 [.6977, .7108]	.4364 [.4281, .4444]
	SigLIP2-B/16@384	.7031 [.6970, .7093]	.4330 [.4254, .4413]
	FashionCLIP 2.0	.6981 [.6920, .7042]	.4275 [.4193, .4358]
	Marqo-FashionSigLIP	**.7221 [.7158, .7282]**	**.4451 [.4371, .4539]**
	GR-Lite	.6511 [.6440, .6580]	.3832 [.3757, .3911]

historical positive split with regenerated clean-style CP/FITB questions. Clean-minus-bridge changes are only −0.0017 to 0.0061 AUC and −0.0023 to 0.0106 FITB, whereas packaged-historical to regenerated-question changes are appreciably larger for several models. Thus a score may increase after moving to the clean protocol because the negative/distractor instrument changed; leakage removal itself is not interpreted as improving representations. Detailed bridge values are in the supplement. Historical results remain comparison-only.

7.3 Prospective replication on frozen IQON3000-Clean (RQ1)

Absolute performance falls on IQON, consistent with a more heterogeneous user-curated corpus and a different negative/distractor population. The separation nonetheless persists: CLIP, SigLIP2, and FashionCLIP all remain clearly above generic DINOv3, which in turn exceeds ResNet50 by 0.0566 AUC (95% CI 0.0508–0.0626) and 0.0494 FITB (0.0416–0.0580). GR-Lite exceeds DINOv3 by 0.0092 AUC (0.0026–0.0155) and 0.0172 FITB (0.0092–0.0255). Each predefined contrast remains supported after correction.

Within the full roster, Marqo-FashionSigLIP has the best point estimate for both tasks: 0.7221 AUC and 0.4451 FITB. It improves over SigLIP2 by 0.0190 AUC (0.0145–0.0234) and 0.0121 FITB (0.0050–0.0190), unlike the null Polyvore contrast. Against CLIP, the best specialized model adds 0.0179 AUC and 0.0087 FITB. FashionCLIP again trails CLIP, by 0.0062 AUC and 0.0089 FITB. The two datasets thus show different benefits from specialization, depending on the released checkpoint and training recipe. Marqo's IQON lead is a checkpoint-specific result with additional provenance uncertainty; the replicated VLM-tier advantage does not rely on it.

7.4 Cross-dataset replication (RQ1)

Figure 1 separates absolute difficulty from relative ordering. CP ranks correlate at Spearman $\rho = 0.964$ and Kendall $\tau = 0.905$; FITB ranks correlate at $\rho = 0.857$ and $\tau = 0.714$. CP and FITB orderings within IQON are identical. ResNet50 is last and DINOv3 next-to-last on both clean datasets; the contrastive VLMs occupy the upper group; GR-Lite lies between generic DINOv3 and that group. The replication supports the hierarchy despite sizeable absolute score shifts.

IQON differences among the upper models are more visible, and Marqo's advantage appears only there. The stable result is the tier-level separation: after excluding both uncertain-provenance checkpoints, SigLIP2 leads Polyvore and CLIP leads IQON, while the evaluated contrastive VLMs remain above ResNet50 and generic DINOv3. The

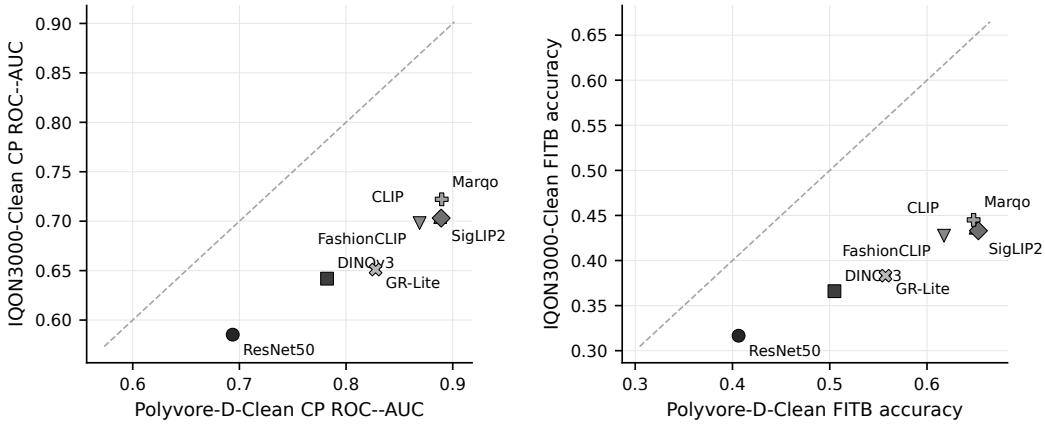


Fig. 1 Representation performance and rank replication across Polyvore-D-Clean and IQON3000-Clean. IQON is the prospective replication on the frozen leakage-audited target; connecting lines show the same frozen representation, not paired examples across corpora.

hierarchy is therefore not peculiar to one set of Polyvore products, one clean split construction, or the inclusion of Marqo and GR-Lite.

7.5 A100 external human/expert-grounded transfer (RQ2)

Figure 2 summarizes the external results. Polyvore-trained representations obtain LAT/AAT accuracies of 0.37/0.35 (ResNet50), 0.39/0.56 (DINOv3), 0.60/0.67 (CLIP), 0.50/0.63 (SigLIP2), 0.58/0.62 (FashionCLIP), 0.62/0.64 (Marqo), and 0.54/0.59 (GR-Lite). IQON-trained probes obtain 0.24/0.31, 0.34/0.40, 0.50/0.56, 0.48/0.63, 0.56/0.57, 0.58/0.63, and 0.47/0.58, respectively. Best-over-ResNet effects range from 0.23 to 0.34 on LAT and are 0.32 on AAT for either training source. These predefined effects remain supported after correction.

Other A100 contrasts are less conclusive at $n = 100$. For example, Polyvore-trained Marqo exceeds SigLIP2 on LAT by 0.12 (0.03–0.21; Holm-adjusted $p = 0.045$), but the corresponding AAT contrast is only 0.01 and the IQON-trained contrasts are not supported. Many intervals among the evaluated VLMs overlap. Rank correlations between clean CP and A100 are positive: for Polyvore, $\rho = 0.786$ with LAT and 0.893 with AAT; for IQON, 0.893 and 0.721. With seven systems and 100 questions, these correlations do not resolve small differences between models. mLAT gives the same broad picture, with best values of 0.501 (Polyvore-trained Marqo) and 0.474 (IQON-trained Marqo). The alternative archive label interpretation affects seven questions and is reported in the supplement; it does not reverse the broad tier-level conclusion. A100 supports the broad transfer result, with limited power to rank closely matched models.

7.6 Task-specific contextual baseline (RQ2)

The image-only OutfitTransformer adaptation reaches 0.8095 ± 0.0020 CP AUC and 0.4655 ± 0.0051 FITB on Polyvore, versus 0.7723 ± 0.0013 and 0.4354 ± 0.0012 on IQON (mean $\pm$ training-seed SD). Its IQON CP result exceeds every frozen-probe result, showing that supervised contextual learning can compensate for a conventional ResNet input when sufficient in-domain labels are available. On IQON FITB, however, it is approximately tied with CLIP and below Marqo's 0.4451. On Polyvore it trails all VLM probes on both tasks and trails DINOv3 on FITB.

External transfer is also mixed. Polyvore-trained baseline means are 0.490 ± 0.051 LAT and 0.340 ± 0.029 AAT; IQON-trained means are 0.418 ± 0.026 and 0.382 ± 0.054. The baseline is thus competitive for IQON CP but does not dominate across task, dataset, or external metric. Because its visual input, optimization, and capacity differ from the frozen probes, this comparison is diagnostic rather than a controlled representation contrast. It is an image-only contextual

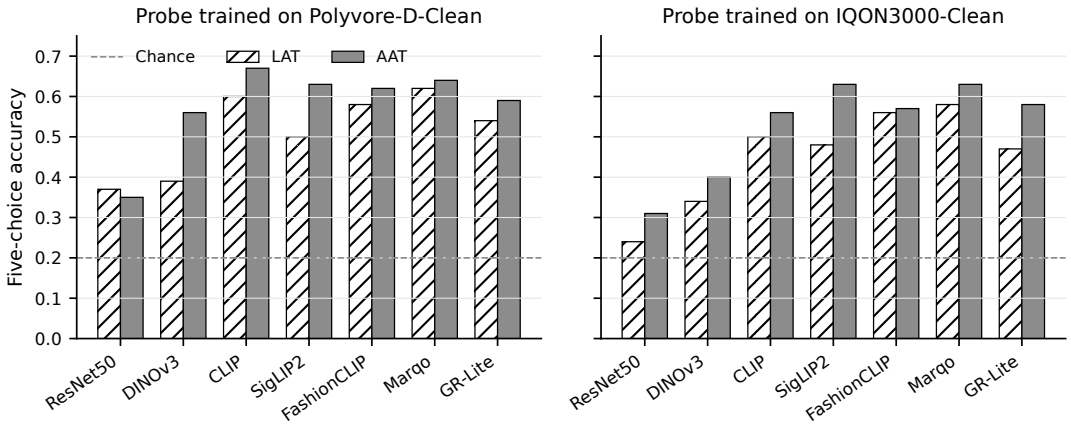


Fig. 2 A100 LAT and AAT accuracy for probes trained on each clean source. Each task has 100 fixed questions; uncertainty is question-bootstrap uncertainty. A100 was never used for fitting or model selection.

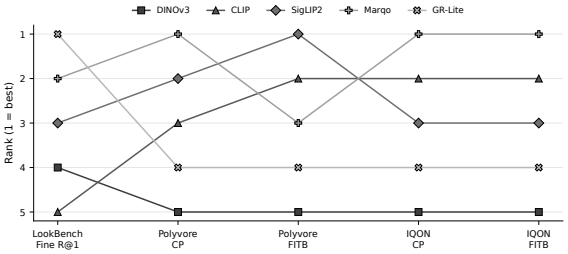


Fig. 3 LookBench Fine R@1 versus clean compatibility for the exact five-model overlap. With only five systems, correlations are descriptive and rank reversals are more informative than a test of zero association.

adaptation, not the canonical published implementation. Published VICTOR, HAT, MEDAL, HON, Text2Outfit, and multimodal results cannot be inserted as like-for-like baselines because their splits, modalities, personalization targets, negatives, or evaluation candidates differ.

7.7 Retrieval versus relational compatibility (RQ2)

LookBench version 3 (revised 13 April 2026) reports Fine Recall@1 for an exact checkpoint overlap of DINOv3 (43.97), CLIP (39.79), SigLIP2 (59.44), Marqo (62.77), and GR-Lite (65.71) [19]. On this subset, Polyvore compatibility correlation is $\rho = 0.20$ for CP and $\rho = -0.20$ for FITB (both nominal $p = 0.747$); both IQON correlations are exactly $\rho = 0$. The ranks exhibit concrete reversals: GR-Lite leads this retrieval subset but is fourth for both IQON compatibility tasks; CLIP has the lowest retrieval score yet ranks second on IQON CP/FITB and near the Polyvore leaders. Figure 3 therefore shows that retrieval rank is not a reliable deterministic compatibility-selection rule in this exact overlap. It does not establish population-level independence: five models are too few, the correlations are descriptive, LookBench measures a different construct, and ResNet50 and FashionCLIP are absent.

7.8 Benchmark calibration (RQ3)

The reliability curves in Fig. 4 show why model choice depends on the calibration measure. Marqo has the lowest Brier score on both Polyvore (0.1322) and IQON (0.2197), consistent with its strong ranking and probability accuracy. ECE_{15}, however, is lowest for ResNet50: 0.0288 on Polyvore and 0.0415 on IQON. Marqo records 0.0350 and 0.0583, and FashionCLIP has the second-largest IQON ECE at 0.0571, below Marqo's 0.0583. These small absolute differences depend on binning; the 10/20-bin analyses retain the broader conclusion that no single model dominates all calibration summaries.

A weak discriminator can place many predictions near the constructed 0.5 base rate and appear locally calibrated, whereas a stronger model may require post-hoc recalibration. Brier score blends refinement and calibration; ECE discretizes reliability; ROC–AUC ignores probability scale. All three probability summaries describe *benchmark calibration* under balanced synthetic CP, not calibration to the unknown real-world prevalence of compatible outfits. We therefore

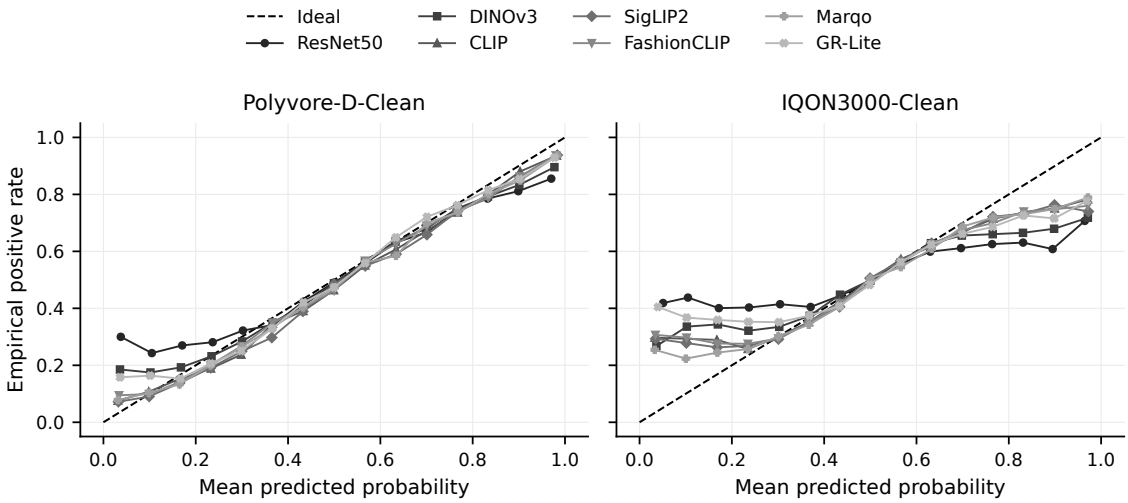


Fig. 4 Reliability diagrams for CP probabilities under the balanced constructed benchmark protocols. ECE uses 15 equal-width bins; 10- and 20-bin sensitivity results are in the supplement. These curves do not estimate real-world compatibility prevalence.

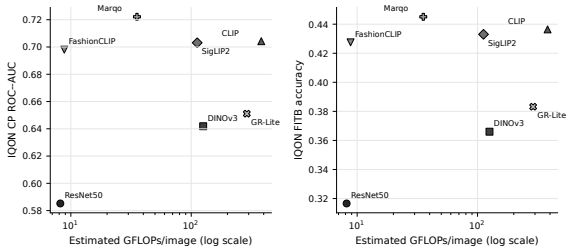


Fig. 5 IQON accuracy versus estimated encoder GFLOPs. Pareto status depends on the selected accuracy, latency, memory, and provenance constraints; it is not intrinsic to a model.

report them jointly and would fit any deployment recalibration on held-out deployment-like data. FITB scores are rankings, not calibrated probabilities.

7.9 Computational efficiency and Pareto analysis (RQ3)

Table 6 reports the measured costs and Fig. 5 relates them to accuracy. Encoder extraction dominates the light downstream pipeline: IQON PCA takes 5.1–12.4 s and logistic fitting 1.7–3.2 s, compared with 409–4,778 s for feature extraction. CLIP is the most expensive measured option (381.9 GFLOPs, 9.31 ms/image, 2.03 GB peak VRAM) despite near-leading accuracy. FashionCLIP is light (8.82 GFLOPs and 401 MB) but less accurate. Marqo combines the best IQON accuracy with 1.91 ms/image and 651 MB, placing it on the full-roster frontier for the measured criteria. Its provenance uncertainty remains a separate cost that a numerical Pareto plot cannot encode. Excluding the two checkpoints with the greatest fashion-data provenance uncertainty changes the frontier. Predictive quality, hardware cost, memory, calibration, and provenance therefore support different model choices.

7.10 Stability and robustness (RQ4)

Figure 6 shows that construction-seed variability is small relative to the main representation gaps. Across seeds, Polyvore CP rank agreement with the primary construction is $\rho = 0.893$–1.000 and FITB agreement is 0.964; IQON ranges from 0.964–1.000 for both tasks. Typical CP standard deviations are 0.0005–0.0021 and FITB standard deviations 0.0013–0.0054. The leading VLMs can exchange adjacent positions, consistent with their overlapping intervals, but the broad family hierarchy survives.

As Fig. 7 shows, native-dimensional features increase IQON CP/FITB for every model: for example, ResNet50 rises from 0.5852/0.3166 to 0.6674/0.3644, and Marqo from 0.7221/0.4451 to 0.7593/0.4663. Yet rank correlations between PCA-256 and native features are $\rho = 0.857$, $\tau = 0.714$ for both tasks. XGBoost over the

Table 6 Same-device encoder efficiency and IQON extraction cost. Latency is repeated batch-64 inference on one RTX 5070 Ti; GFLOPs are PyTorch estimates. IQON extraction used batch 32 over 434,089 item images.

Representation	GFLOPs	ms/image	images/s	VRAM (MB)	IQON time (s)	IQON images/s
ResNet50	8.17	1.976	506.1	764	408.7	1,062.1
DINOv3 ViT-L/16	125.68	3.145	318.0	885	1,578.7	275.0
CLIP ViT-L/14@336	381.92	9.313	107.4	2,029	4,777.6	90.9
SigLIP2-B/16@384	112.13	5.882	170.0	1,443	2,612.2	166.2
FashionCLIP 2.0	8.82	2.282	438.2	401	717.7	604.8
Marqo-FashionSigLIP	35.42	1.907	524.3	651	655.9	661.8
GR-Lite	289.62	6.828	146.5	1,687	2,832.7	153.2

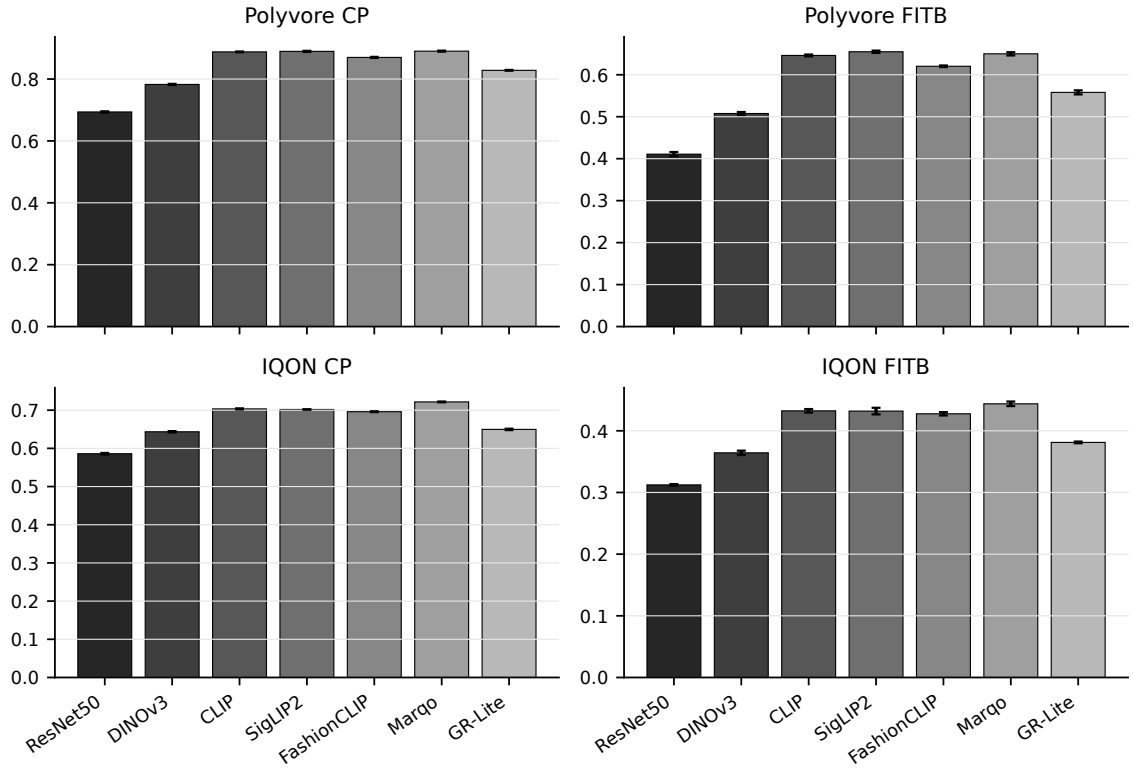


Fig. 6 Construction-seed stability when item partitions are held fixed but CP corruptions and FITB distractors are regenerated for seeds 42–46. Error bars summarize construction-seed variation, not sampling uncertainty.

same IQON outfit features similarly preserves CP ranks ($\rho = 0.964$, $\tau = 0.905$) and exactly preserves FITB ranks, although absolute FITB often decreases. Together with the five construction seeds, these analyses show that the main family-level hierarchy is not specific to PCA width, one linear decision boundary, or one generated-question seed.

The tier-level result persists after exclusion of Marqo and GR-Lite, the two checkpoints with the greatest fashion-data provenance uncertainty. SigLIP2 then leads Polyvore and CLIP leads IQON, while FashionCLIP still trails CLIP. This provenance-restricted analysis supports the VLM-tier advantage and the recipe-dependent specialization result.

The compatibility-label efficiency curves are shown in Fig. 8. At 10% of Polyvore labelled outfits, CLIP and Marqo already reach approximately 0.863 CP AUC, while ResNet50 reaches 0.642 and DINOv3 0.743. More labels improve every model, but do not close the representation gap. Because PCA continues to see every train-split item embedding, this isolates labelled-outfit

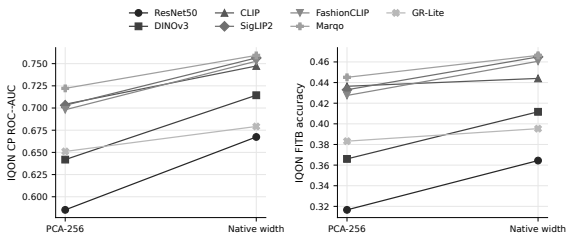


Fig. 7 IQON PCA-256 versus native-dimensional robustness. Native features improve absolute values for all representations; rank order remains similar.

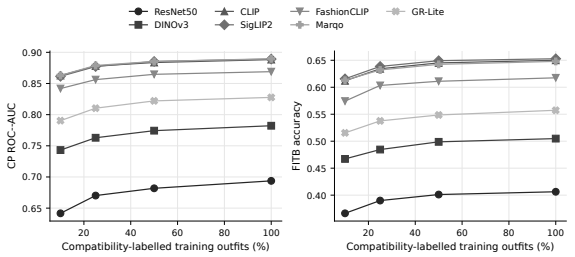


Fig. 8 Polyvore-D-Clean compatibility-label efficiency. Fractions reduce labelled training outfits; frozen encoders and PCA access to all train-split item embeddings remain unchanged.

demand rather than generic low-data learning or few-shot encoder adaptation.

Outfit-length and failure analyses identify remaining structure. Mean CP AUC across models rises from 0.721 for two-item to 0.888 for six-or-more-item Polyvore outfits, and from 0.602 to 0.758 on IQON. Mean FITB rises more modestly (0.520 to 0.584; 0.355 to 0.442). Longer outfits supply more pair evidence but also change the example distribution, so the trend is descriptive. Unanimous errors are relatively rare compared with model disagreements: 463 versus 19,108 of 30,284 Polyvore CP examples and 1,523 versus 39,854 of 49,600 IQON CP examples; FITB has 965/12,089 of 15,142 and 3,497/20,200 of 24,800, respectively. The large disagreement pool suggests that representations expose meaningfully different relational cues even where aggregate ranks are stable.

8 Discussion

8.1 Answer to RQ1: controlled transfer and replication

Our results establish a stable tier-level advantage for the evaluated contrastive VLM representations over ResNet50 and generic DINOv3 under the shared relational probe. The broad ordering replicates prospectively on the frozen IQON3000-Clean protocol despite lower absolute scores: cross-dataset rank correlations are 0.964 for CP and 0.857 for FITB. Removing Marqo-FashionSigLIP and GR-Lite leaves the same family-level conclusion. Because the released checkpoints differ as complete packages, the comparison directly supports checkpoint selection, while the matched specialization contrasts distinguish family-level strength from recipe-specific gains.

8.2 Answer to RQ2: construct and modelling boundaries

The hierarchy extends beyond the synthetic clean questions at tier level. Unchanged A100 questions broadly corroborate the evaluated VLM advantage, while their sample size precludes reliable ordering of close VLMs. The contextual baseline leads on IQON CP, but its performance varies across datasets, tasks, and external metrics. Fashion specialization is checkpoint- and recipe-dependent: FashionCLIP trails CLIP, whereas Marqo improves on IQON. Exact-overlap LookBench rank reversals further establish that a fashion-retrieval leaderboard is not a reliable compatibility-selection rule for the matched checkpoints.

8.3 Answer to RQ3: operational selection axes

Marqo leads IQON discrimination and Brier score and is efficient on the measured GPU, but its fashion-data provenance is less certain and it does not minimize ECE. ResNet50 has the lowest benchmark ECE and low cost but weak discrimination. CLIP is predictively strong and computationally expensive; FashionCLIP is compact but

does not lead compatibility. These findings concern balanced-protocol calibration and one hardware/software stack. Model choice should therefore account for constraints on error rates, probability quality, latency, memory, reproducibility, and data governance.

8.4 Answer to RQ4: validity and robustness

The main family hierarchy survives exact-collision audits, five regenerated question seeds, native-width features, XGBoost, uncertain-provenance exclusion, compatibility-label fractions, and conservative IQON item-component resampling. The historical bridge shows that apparent historical-to-clean score increases chiefly reflect regenerated questions rather than split cleaning. The IQON construction makes the trade-off between split integrity and source-population coverage measurable: 98.74% of source users remain represented and category divergence is 0.0016 bits, while retained outfits are shorter and use less frequently repeated items. Predefined contrast directions are preserved under conservative item-component resampling. None of these tested alternatives reverses the broad ordering of representation families.

8.5 Practical implications for machine-vision systems

For fashion-vision research and deployment, an encoder should be selected on relational compatibility evidence rather than a fashion-retrieval leaderboard: the exact LookBench overlap contains consequential rank reversals. Fashion-domain pretraining is likewise not a sufficient selection criterion, because its effect depends on the released recipe. Strong frozen contrastive VLM features can support competitive compatibility prediction without encoder fine-tuning, making frozen transfer a practical baseline when training cost, update frequency, or data access constrain end-to-end adaptation.

Accuracy should not be the sole operational criterion. Benchmark calibration, latency, throughput, memory, and training-data provenance change the preferred checkpoint, and these constraints should be reported alongside discrimination. At the benchmark level, cross-split item identifiers and exact images must be audited together, connected duplicate structure must be respected during splitting, and generated CP negatives and FITB distractors must be checked against known positives and within-question duplication. These practices make machine-vision comparisons more interpretable and reduce the risk that an apparent model gain is instead a property of data leakage or question construction.

9 Threats to validity and limitations

Construct validity. CP negatives and FITB distractors are synthetic, although they are same-category, within-split, deterministically generated, collision-audited, and tested across five construction seeds. They approximate plausible confusions rather than naturally occurring disliked outfits; harder similarity-mined negatives remain a useful extension. Compatibility itself is culturally and contextually dependent, and image-only probes omit personal preference, garment fit, price, brand, and text. A100 improves human grounding but remains small. Its expert/human labels and facet partitions do not exhaust aesthetic judgments.

Internal validity. Exact identifiers, encoded files, decoded pixels, and duplicate outfits are controlled across the leakage-audited splits, and generated questions have no audited exact positive collisions. Transformed or perceptually near-duplicate images cannot be exhaustively excluded. IQON hypergraph packing retains 41.62% of valid outfits; it preserves nearly all users and the category mixture but shifts outfit length and item repetition, so it is not assumed to represent every valid source outfit. IQON is item/image-disjoint but not user-disjoint and therefore is not a new-user generalization benchmark. User-cluster bootstrap addresses one dependence source; item-component sensitivity preserves the main contrasts but is dominated by one large component.

Representation attribution and upstream provenance. The seven checkpoints differ in model size, input resolution, architecture, data, objective, and post-processing, so the results compare released representations as packages rather than causally isolating one pretraining factor. The split audits do not establish whether CLIP, SigLIP2,

DINOv3, FashionCLIP, Marqo-FashionSigLIP, GR-Lite, or visually related upstream content ever encountered Polyvore or IQON during web-scale pretraining. Marqo-FashionSigLIP and GR-Lite carry the greatest additional uncertainty because their fashion data are not item-wise disclosed. Excluding both preserves the tier-level result but does not certify the remaining web-pretrained models.

External validity. Two clean fashion corpora and A100 do not cover all cultures, markets, product photography, or temporal trends. Historical Polyvore is included only for comparison. The LookBench analysis contains exactly five overlapping models and mixes an external retrieval metric with compatibility metrics; its correlations are descriptive. The efficiency study uses one recent consumer GPU, one software stack, fixed batch settings, and estimated FLOPs. Other accelerators, quantization, compilation, or serving loads could alter the frontier.

Probe and baseline validity. The task-specific baseline is an image-only contextual adaptation built from the published OutfitTransformer design rather than an exact reproduction of its canonical multimodal training. Native-width and XGBoost sensitivity analyses cover plausible downstream alternatives but do not exhaust all possible relational probes.

Statistical validity. Group-aware paired bootstrap preserves important dependence, and Holm correction limits the predefined family, but confidence intervals still inherit assumptions about the resampling unit. Construction-seed analysis covers five seeds; A100 intervals are wide; rank correlations over seven or five systems have low resolution. The retained-subset and collision diagnostics were post-audit validity analyses, not preregistered tests. Null or overlapping results are reported as inconclusive rather than evidence of equivalence. CP calibration is conditional on the constructed 50:50 protocol and cannot be transported to real-world prevalence without new calibration data.

10 Reproducibility and data/code availability

The frozen protocol hash begins `96ffd69883c381f8` and the IQON split-manifest hash begins `f5081ec5d2ea0262`; the supplement gives both complete SHA-256 values. The protocol specification and hash record are included with the experiment configurations. The public repository at https://github.com/alifarzadjamali/fashion-compatibility-benchmark contains the extraction and evaluation code, locked environment specification, checkpoint revisions, experiment configurations, split builders, and statistical analyses. Its README documents the execution sequence and reports reference results. The recorded run environment is Windows 11, Python 3.12.13, PyTorch 2.11.0+cu128, Transformers 4.57.6, scikit-learn 1.9.1, CUDA 12.8, and an RTX 5070 Ti.

Reproduction requires acquiring Polyvore, IQON3000, and A100 under their source terms, obtaining pretrained weights from the original providers, and running the extraction and evaluation pipeline. Source images, weights, cached embeddings, fitted models, raw predictions, generated reports, and manuscript files are excluded from the Git repository. The scripts generate local audit records, result tables, and figures. The manuscript and supplement report the frozen experimental results; hardware and numerical-library differences can affect regenerated values.

11 Conclusion

Under a shared relational probe, the evaluated contrastive VLM representations outperform ResNet50 and generic DINOv3 on both leakage-audited corpora. A100 supports this broad ordering, which also persists after excluding the two checkpoints with the greatest provenance uncertainty. Fashion specialization depends on the checkpoint and training recipe, and retrieval ranks do not reliably predict compatibility ranks. Calibration and computational cost further affect checkpoint choice. The historical question bridge, collision audits, five construction seeds, feature-width and nonlinear-learner checks, label-fraction analysis, and dependence-aware sensitivity analyses leave the main ordering intact. The released code and frozen configurations support reproduction of these comparisons and their extension to other representations.

Supplementary information. Online Resource 1 supplies additional protocol, robustness, calibration, provenance, subgroup, and statistical evidence.

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Statements and Declarations

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Data availability. Code for generating audit summaries, derived result tables, and statistical outputs, together with the frozen protocol specification, is available at https://github.com/alifarzadjamali/fashion-compatibility-benchmark. Original Polyvore, IQON3000, and A100 images are not redistributed and must be obtained from their respective sources subject to their terms. Raw per-example predictions and cached embeddings are not included in the Git repository.

Code availability. The evaluation code, frozen configuration, protocol hashes, checkpoint revisions, and reproduction instructions are publicly available at https://github.com/alifarzadjamali/fashion-compatibility-benchmark.

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